

Program 5

STEP 1 : Create Project Director

Step 2 : Create Dockerfile

Step 3 : Create docker-compose.yaml

```
(1RV24MC114_VENKATESHAHD@kali) - [~/mca/devops/lab/lab5]
$ ls
docker-compose.yaml  Dockerfile
(1RV24MC114_VENKATESHAHD@kali) - [~/mca/devops/lab/lab5]
$ █
```

Step 4 : Build & Run Jenkins

```

(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab5]
$ docker compose up --build
WARN[0000] /home/1RV24MC114_VENKATESHAHD/mca/devops/lab/lab5/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 0.1s (8/8) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 550B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 144B 0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lts-jdk11 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:lts-jdk11 0.0s
=> CACHED [2/2] RUN apt-get update && apt-get install -y docker.io 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:2a4af79e84e1f5ce38560b2d27a75f294ff4a3baa3bef92f924d6cbd0b3bdeac 0.0s
=> => naming to docker.io/library/lab5-jenkins 0.0s
=> resolving provenance for metadata file 0.0s
[+] up 2/2
✓ Image lab5-jenkins Built 0.2s
✓ Container jenkins-ci Created 0.0s
Attaching to jenkins-ci
jenkins-ci | Running from: /usr/share/jenkins/jenkins.war
jenkins-ci | webroot: /var/jenkins_home/war
jenkins-ci | 2026-01-04 15:51:48.609+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
jenkins-ci | 2026-01-04 15:51:48.658+0000 [id=1] WARNING o.e.j.s.handler.ContextHandler#setContextPath: Empty contextPath
jenkins-ci | 2026-01-04 15:51:48.714+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-10.0.24; built: 2024-08-26T17:58:21.070Z; git: d5384207795da96fad32db8ea8d26b69955bcc03; jvm 11.0.24+8
jenkins-ci | 2026-01-04 15:51:48.938+0000 [id=1] INFO o.e.j.w.StandardDescriptorProcessor#visitServlet: No JSP Support for /, did not find org.eclipse.jetty.jsp.JettyJspServlet
jenkins-ci | 2026-01-04 15:51:48.980+0000 [id=1] INFO o.e.j.s.s.DefaultSessionIdManager#doStart: Session workerName=node0
jenkins-ci | 2026-01-04 15:51:49.327+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkins_home found at: EnvVars.masterEnvVars.get("JENKINS_HOME")
jenkins-ci | 2026-01-04 15:51:49.429+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started w.03ff57625{Jenkins v2.462.3,/,file:///var/jenkins_home/war/,AVAILABLE}{/var/jenkins_home/war}
jenkins-ci | 2026-01-04 15:51:49.437+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started ServerConnector@55740540{HTTP/1.1, (http/1.1)}{0.0.0.0:8080}
jenkins-ci | 2026-01-04 15:51:49.446+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started Server@73ee04c8{STARTING}[10.0.24,sto=0] @1201ms
jenkins-ci | 2026-01-04 15:51:49.448+0000 [id=25] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
jenkins-ci | 2026-01-04 15:51:49.609+0000 [id=32] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
jenkins-ci | 2026-01-04 15:51:49.650+0000 [id=52] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
jenkins-ci | 2026-01-04 15:51:50.193+0000 [id=44] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
jenkins-ci | 2026-01-04 15:51:50.199+0000 [id=45] INFO jenkins.InitReactorRunner$1#onAttained: Started all

```

```

jenkins-ci | 2026-01-04 15:51:49.650+0000 [id=52] INFO jenkins.InitReactorRunner$1#onAttained: Listed all
jenkins-ci | 2026-01-04 15:51:50.193+0000 [id=44] INFO jenkins.InitReactorRunner$1#onAttained: Prepared al
jenkins-ci | 2026-01-04 15:51:50.199+0000 [id=45] INFO jenkins.InitReactorRunner$1#onAttained: Started all
jenkins-ci | 2026-01-04 15:51:50.205+0000 [id=48] INFO jenkins.InitReactorRunner$1#onAttained: Augmented a
jenkins-ci | 2026-01-04 15:51:50.382+0000 [id=36] INFO jenkins.InitReactorRunner$1#onAttained: System conf
jenkins-ci | 2026-01-04 15:51:50.382+0000 [id=36] INFO jenkins.InitReactorRunner$1#onAttained: System conf
jenkins-ci | 2026-01-04 15:51:50.382+0000 [id=30] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all
jenkins-ci | 2026-01-04 15:51:50.383+0000 [id=37] INFO jenkins.InitReactorRunner$1#onAttained: Configurati
jenkins-ci | 2026-01-04 15:51:50.403+0000 [id=67] INFO hudson.util.Retrier#start: Attempt #1 to do the act
jenkins-ci | 2026-01-04 15:51:50.433+0000 [id=41] INFO jenkins.install.SetupWizard#init:
jenkins-ci |
jenkins-ci | *****
jenkins-ci | *****
jenkins-ci | *****
jenkins-ci | *****
jenkins-ci | Jenkins initial setup is required. An admin user has been created and a password generated.
jenkins-ci | Please use the following password to proceed to installation:
jenkins-ci |
jenkins-ci | 241f253476a04093b21a8122ebc98aa0
jenkins-ci |
jenkins-ci | This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
jenkins-ci |
jenkins-ci | *****
jenkins-ci | *****
jenkins-ci | *****
jenkins-ci | *****
jenkins-ci | WARNING: An illegal reflective access operation has occurred
jenkins-ci | WARNING: Illegal reflective access by org.codehaus.groovy.vmplugin.v7.Java7$1 (file:/var/jenkins_home
jenkins-ci | /war/WEB-INF/lib/groovy-all-2.4.21.jar) to constructor java.lang.invoke.MethodHandles$Lookup(java.lang.Class,int)
jenkins-ci | WARNING: Please consider reporting this to the maintainers of org.codehaus.groovy.vmplugin.v7.Java7$1
jenkins-ci | WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
jenkins-ci | WARNING: All illegal access operations will be denied in a future release
jenkins-ci | 2026-01-04 15:51:54.926+0000 [id=41] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
jenkins-ci | 2026-01-04 15:51:54.980+0000 [id=24] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
jenkins-ci | 2026-01-04 15:51:56.778+0000 [id=67] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
jenkins-ci | 2026-01-04 15:51:56.778+0000 [id=67] INFO hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1

```

STEP 5 :Confirm Jenkins running

STEP 6 : GET Jenkins admin key

STEP 7 : Verify Docker Inside Jenkins

```

1RV24MC114_VENKATESHAHD@kali)~[~/mca/devops/Lab/Lab5]
$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
36cbca1f3d54   lab5-jenkins   "/usr/bin/tini -- /u..." 2 minutes ago   Up 2 minutes   0.0.0.0:8080->8080/tcp, :::8080->8080/tcp, 50000/tcp   jenkins-ci
1RV24MC114_VENKATESHAHD@kali)~[~/mca/devops/Lab/Lab5]
$

```

Step 8 : Access Jenkins UI

